# CS3205+CS3210 — Introduction to Computer Networks (+ lab)

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|---|---|
| Tier | A |
| Phase (order) | 4 |
| Do this first | CS2705, CS3500 helps |
| Estimated effort | 12 weeks |
| Unlocks | CS6500, CS6048, CS6847 |

## Roadmap

1. Layered model, application layer (HTTP, DNS), sockets.
2. Transport: TCP, UDP, congestion control.
3. Network layer: IP, routing; link layer: Ethernet, ARP.
4. Hands-on: Wireshark captures, and a socket-based client/server.

## Video courses (all links checked on YouTube on 2026-10-22)

### 🎓 Main course: follow it start to end

- Computer Networks and Internet Protocol (61 videos). NPTEL *Computer Networks and Internet Protocol* (Soumya Kanti Ghosh & Sandip Chakraborty, IIT Kharagpur): has socket-programming demos.

### 🎥 Other full lecture series (pick one if the main one doesn't click, or for topics it skips)

- Computer Networks (34 videos). IIT Madras BS *Computer Networks*.
- Computer Networking: A Top-Down Approach - Jim Kurose (47 videos). Jim Kurose's own lectures for *Computer Networking: A Top-Down Approach* (collected by a viewer): matches the course textbook chapter by chapter.

### ⚡ Short videos for intuition and revision

- Computer Networks (100+ videos). Neso Academy *Computer Networks*: subnetting/addressing drills.

## Study plan (week by week)

Lecture numbers are positions in the main playlist above. Click a number to open that video.

| Week | Lectures | What to do |
|---|---|---|
| 1 | 2–6 | History, circuit vs packet switching, the protocol stack. Install Wireshark and capture your own traffic. |

| Week | Lectures | What to do |
|---|---|---|
| 2 | 16–20 | Application layer: DNS, FTP, HTTP, SMTP. Use dig/curl -v; write a tiny HTTP server in Python with raw sockets. |
| 3-4 | 7–15, 21–24 | Transport: reliability, sliding window, TCP connection/flow/congestion control, UDP. (The playlist order is shuffled; follow the lecture numbers in the titles.) Do the Kurose Wireshark TCP lab. |
| 5 | 25–26 | Socket programming. Build a multi-client chat server in C or Python. |
| 6-7 | 27–42 | Network layer: IPv4/CIDR/NAT, IPv6, QoS, routing tables, intra/inter-domain routing, BGP, routers. Subnetting exercises; simulate Dijkstra/distance-vector. |
| 8 | 43–46 | Software-defined networking. Try Mininet. |
| 9 | 47–56, 59 | Data link and physical layer: Ethernet, error control, ARP/DHCP, VLANs, wireless LANs. |
| 10 | 57–58, 60 | Network security overview. Then TLS from Kurose ch. 8. |

▼ **All lectures in *Computer Networks and Internet Protocol* (61 videos)**

1. Prof Soumya Kanti Ghosh & Prof Sandip Chakraborty
2. Lecture 1:Introduction to Computer Networks – A brief history
3. Lecture 2:Data Networks – from Circuit Switching Network to Packet Switching Network
4. Lecture 3:Network Protocol Stack
5. Lecture 4:Services at the Different Layers of the Protocol Stack
6. Lecture 5:Application Layer I – Different Protocols at the Application Layer
7. Lecture 11:Transport Layer I – Services
8. Lecture 12:Transport Layer II - Connection
9. Lecture 14:Transport Layer IV – Reliability
10. Lecture 16:Transport Layer Performance
11. Lecture 17:Buffer Management and Congestion Control
12. Lecture 18:Transport Layer Primitives
13. Lecture 19:Transmission Control Protocol I – Basics
14. Lecture 20:Transmission Control Protocol II – Connections
15. Lecture 13:Transport Layer II - Connection (Contd.)
16. Lecture 6:Application Layer II – Domain Name Systems
17. Lecture 7:Application Layer III – File Transfer (FTP)
18. Lecture 8:Application Layer IV – Hypertext Transfer Protocol (HTTP), HTML, Telnet Part-I
19. Lecture 9:Application Layer IV – Hypertext Transfer Protocol (HTTP), HTML, Telnet Part-II
20. Lecture 10:Application Layer V – Internet Mail Transfer (SMTP), Network Management (SNMP)
21. Lecture 15: Transport Layer V – Sliding Window Protocols
22. Lecture 21: Transmission Control Protocol III – Flow Control
23. Lecture 22: Transmission Control Protocol IV – Congestion Control
24. Lecture 23: User Datagram Protocol

25. Lecture 24 : Socket Programming – I
26. Lecture 25: Socket Programming – II
27. Lecture 26: Network Layer I – Introduction
28. Lecture 27: IP Addressing (IPv4) I – Classful addressing
29. Lecture 28: IP Addressing (IPv4) II - CIDR
30. Lecture 29: IP Addressing (IPv4) III – Network Address Translation (NAT)
31. Lecture 30: IPv6 Addressing
32. Lecture 31: Internet QoS - I (What is QoS)
33. Lecture 32: Internet QoS - II (Basic QoS Architecture)
34. Lecture 33: Internet QoS - III (Traffic Policing and Traffic Shaping)
35. Lecture 34: Internet QoS - IV (Traffic Scheduling)
36. Lecture 35: Internet QoS - V (Integrated and Differentiated Service Architecture)
37. Lecture 36: IP Routing Table
38. Lecture 37: Routing in the Internet I – Intra-domain routing
39. Lecture 38: Routing in the Internet II - Routing protocols
40. Lecture 39: Routing in the Internet III – Inter-domain Routing
41. Lecture 40: Routing in the Internet IV – Border Gateway Protocol
42. Lecture 41 : IP Routers
43. Lecture 42 : IP Routers Demo
44. Lecture 43 : Software Defined Networking - I (Basics)
45. Lecture 44 : Software Defined Networking - II (Open Flow)
46. Lecture 45 : Software Defined Networking - III (Demo)
47. Lecture 46: Data Link Layer - Overview
48. Lecture 47: Data Link Layer - Basic Concept
49. Lecture 48: Data Link Layer - Ethernet
50. Lecture 49: Data Link Layer - Ethernet (Contd.)
51. Lecture 50: Data Link Layer - Flow and Error Control
52. Lecture 51: ARP-RAPP-BOOTP-DHCP
53. Lecture 52: ARP-RAPP-BOOTP-DHCP(Contd.)
54. Lecture 54: Wireless LANs
55. Lecture 53: Connectiing LANs,VLAN
56. Lecture 56:Layer 1: Physical Layer-II
57. Lecture 58:Network Security-Overview
58. Lecture 59:Network Security-III(TCP?IPSecurity)
59. Lecture 55: Layer 1: Physical Layer
60. Lecture 60: Network Security-III(TCP/IP Security)
61. mod12Lec57

## Books, notes and websites

- Book: Kurose & Ross, *Computer Networking: A Top-Down Approach* (listed as the course text) + its free Wireshark labs
- ✅ NPTEL *Computer Networks* (Hema A Murthy, IIT Madras) — nptel.ac.in/courses/106106091; alternatives: IIT Kharagpur 106105081 (Sujoy Ghosh), 106105080 (Ajit Pal)
- Stanford CS144 *Introduction to Computer Networking*
- Beej's Guide to Network Programming — beej.us/guide/bgnet
- Book: Peterson & Davie, *Computer Networks: A Systems Approach* (free online)

## Practice / proof of understanding

Write a TCP echo server and a multi-client chat server in C; capture and annotate a TCP handshake and an HTTP request in Wireshark.

## What your IITM course sheet says

### From the PDF: CS3205

**Official description:** Concepts and implementation of computer networks; architecture, protocol layers, inter-networking and addressing; network application development.

**Official syllabus (trimmed):**

> This course builds a practical and analytical foundation in computer networking, from layered Internet design and transport behavior to hands-on packet capture, measurement, debugging, and socket-based application development.

**Official textbooks:** 1. (Any one of these will suffice) 2. Kurose and Ross, "Computer Networking - A top-down approach", Seventh Edition, Pearson, 2017. 3. Andrew S. Tanenbaum, "Computer Networks", Fifth Edition, Pearson Education India, 2013. 4. Peterson and Davie, "Computer Networks, A Systems Approach", 5th ed., Elsevier, 2011.

**Official references:** 1. Ying-Dar Liu, Ren-Hung Hwang, Fred Baker, "Computer Networks: An Open Source Approach", McGraw-Hill, 2011. 2. W. Richard Stevens, Bill Fenner and Andrew Rudoff, "Unix Network Programming", Volumes 1 and 2, Third Edition, Addison-Wesley Professional, 2003. 3. Michael Donahoo, Ken Calvert, Pocket Guide to TCP/IP Socket Programming in C, Morgan Kaufmann Series in Networking, 2000.

### From the PDF: CS3210

**Official description:** Configuration of networking in Linux using ifconfig, route, bind, etc.; configuration of firewall and masquerading in Linux; network trouble-shooting and performance monitoring using netstat, ping, tcpdump, etc. Configuration and performance measurement of commonly-used Linux servers such as E-Mail (sendmail, pop3/imap) and Web (Apache). Socket programming - TCP and UDP, peer-to-peer applications; reliable communications using unreliable datagrams; client-server using RPC; concurrent servers using threads or processes.

**Official textbooks:** NIL

**Official references:** 1. "Linux Network Administrators Guide": http://tldp.org/LDP/nag2/index.html. 2. W.R. Stevens, "Unix Network Programming, Vol 1", 2nd ed., Prentice-Hall Inc., 1998.